Gullak: progress report

Adaptive, multilingual financial literacy for Grades 6–12 with play money. 17 releases, all built and tested in CI.

1. Current approach and changes since M1

The M1 architecture holds: a responsive PWA over pure-TypeScript engine, content-as-data, adaptive models trained in Colab, cached real-data adapters and a Supabase backend behind one data-access layer. Changes since M1:

- **One sim engine, two modes.** Life sim and market sim merged. *Practice* replays real history with life events and a reveal (flagship: March 2020 + a hospital bill); *Live* paper-trades current prices and is scored on habits, never returns.
- **Storyline as the spine.** *Nani's Gullak*: three seasons (one per grade group), a companion and recurring cast, no fixed years in the story, then an "FYI, this really happened" reveal with real Indian data.
- **Reach over games.** Take-home missions make the learner the family's money teacher (households reached is a metric); designed for shared phones and no data; habit-forming, never addictive.
- **Onboarding simplified.** Grade group → current experience (only a starting prior) → optional format ("Mix it up" by default). Evidence, not self-report, drives mastery.
- **Sarvam** replaces Bhashini for English → Hindi / Telugu, with human review.

2. What is built (live at gullak.abgupta.me)

Area	Status
Curriculum as data	33-topic prerequisite graph across the four pillars, tagged by grade group; validated at load (no cycles, no unknown prerequisites). Five demo topics have reviewed-draft curriculum briefs; 61 mutual-fund categories catalogued for the investing pillar.
Onboarding	Phone and desktop flow in en / hi / te; starting cash scales by grade group and experience (₹2,000–₹20,000).
Engine	Compound interest, FD, RD, SIP, goal inverse, unlock rules, grade-aware topic availability, preference-first lesson format picker.
Backend	Verified-session API: onboarding, lesson progress, server-graded quizzes, virtual FD investment, portfolio valuation, activity/episode runs with resumable bookmarks, pinned content hashes and idempotent writes. Row-level security on every learner table; database enforces `is_virtual`.
Real data	AMFI NAV adapter with cache + snapshot fallback; committed NIFTY 50 index-fund history since 2019 (covers March 2020); dated FD-rate snapshot.
Learning UI	Lesson renderer (8-step loop), spot & sort, branching story choices, money-term decoder, paper portfolio, Practice simulation and the counterfactual "rewind" comparison, all localisable.
AI + adaptivity	Gemini → Groq tutor client with advice guardrails (en/hi/te) and grade-aware prompt; offline curriculum-brief validator; BKT input preparation notebook on synthetic data.

3. Experiments and results

- **Format picker:** deterministic tests over every grade group × preference set show explicit preferences are honoured in order, "Mix it up" rotates formats, quizzes are never chosen as lessons, and unavailable formats fall back to authored order with an explicit reason.
- **Knowledge tracing (BKT):** built the version-pinned nine-field quiz-answer contract and a synthetic learner generator with deterministic fixtures and validation tests. Fitting is the next step; parameters fitted on synthetic data will never be used as the production mastery model.
- **Real-data replay:** the committed index-fund series reproduces the March 2020 fall and recovery that the flagship Practice episode needs; the AMFI parser handles both published column layouts.
- **Backend concurrency:** local PostgreSQL tests show retried requests with the same idempotency key apply once, stale revisions are rejected, and two learners on one shared device never see each other's progress.

4. Progress on testing

Layer	Done

Every push and PR	CI runs commit rules, Biome lint, strict typecheck, ~300 Vitest unit tests, production build and Playwright e2e on phone + desktop viewports, including on teammates' forks. A red CI is never merged.
Finance engine	Normal, edge and invalid inputs for every formula; integer paise throughout.
Content	Topic graph, topics, briefs, activities, episodes and snapshots schema-validated in CI; every locale must have identical message keys.
Backend	Strict Zod contracts on every route; RLS isolation, idempotency, rollback and concurrency tests against local Postgres.
Safety	Guardrails screen tutor input and output for advice in en/hi/te; the type system cannot represent real money.
UI	Component tests for lesson, spot & sort, story, portfolio, simulation and rewind flows, including that no screen says "recommended" about a product.

5. Plan for the remaining time (to 11:00)

- **Wire the core loop end to end:** onboarding → lesson → quiz → unlock → spot & sort / story → Practice episode, connecting the new UI to the backend APIs.
- **Server-graded activity answers** and run completion (contract agreed), feeding evidence to mastery.
- **Practice engine:** monthly budget, life events, FD/RD early exit, selling at the day's price; flagship March 2020 episode with the side-by-side reveal.
- **Adaptivity:** fit BKT in Colab, export parameters as JSON, run in the engine; show "Why this?" on recommendations.
- **Hindi and Telugu** for the demo topics via Sarvam; Season 1 storyline episodes; take-home mission with a parent check.
- **Polish and freeze:** design pass on every screen, low-end Android performance and offline checks, guardrail review, e2e for the full loop, final report and demo video.